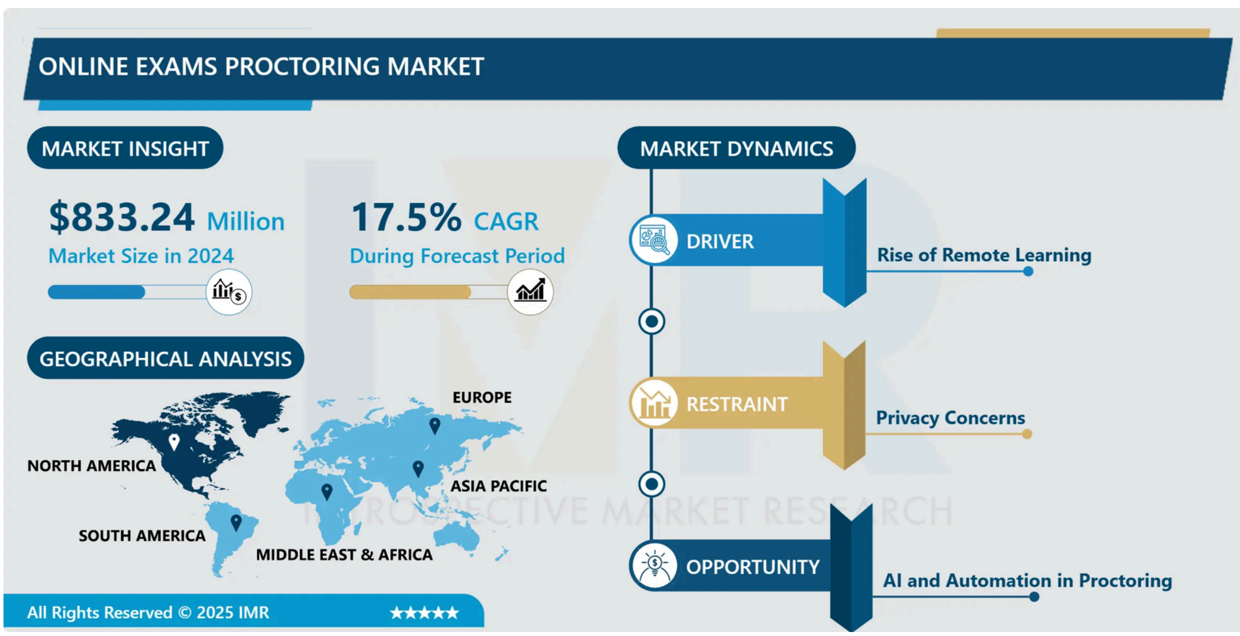
Note: It's a rough document stating the research gap that I am focused on in my FYP. Don't consider it a review paper or some document to be considered for the evaluation criteria.

What my FYP is?
The machine learning used is to fulfil a scope need, not the main focus of my FYP.

The main thing is the research part that I am doing.

Why am I doing research only on this topic:

- See this link: [Online Exams Proctoring Market Trends and Forecast Analysis]
- See this photo taken from this webpage



Research on cheating detection in exam-taking softwares is a major thing, at least after COVID, the rise of remote education and the rapid-growing applications of AI in every field.

Usage of virtual machines is a major cheating tactic: [7 Cheating Tactics in Modern Day Online Exams: How Proctoring Solutions are Fighting Back | Proctortrack]

Exam Taking Software -> Cheating Detection -> Virtual Machines

What I am trying to prove:

1) There has been literally no research on stealth virtual machine detection in exam-taking applications.
2) Little to no research on normal virtual machine detection. Concerns have been raised on bypassing these detection methods.

I am only working on one aspect (virtual machines detection) in this area (Cheating through Online Exam-Taking Softwares). This is a quite a research gap that I have found after working on this topic for a while. See below.

All new research in this domain is only taking POV of ML/AI in it. **Not even a mention of Virtual Machines. That's the research gap that I found and that I am working on.**

- Intelligent Offline Exam Monitoring System for Identifying Suspicious Behavior of the Student (June 2025)
  - Focuses only on AI again.
  - Link: [Intelligent Offline Exam Monitoring System for Identifying Suspicious Behavior of the Student]

- A machine learning assistant for detecting fraudulent activities in synchronous online programming exams (September 2025) (PeerJ Computer Science)
  - Link: [A machine learning assistant for detecting fraudulent activities in synchronous online programming exams [PeerJ]]

Take a look at few of these latest **peer-reviewed review papers** from 2023, 2024, 2025 and even one from 2026 published on 13 February, 2026.

After reading them, you can assure that

1) All research on cheating detection focuses on AI, AI, AI, AI, AI.
2) Not even a single mention of virtual machines in these latest review papers.
3) Remember, they are peer-reviewed, published in quality journals.

- Ensuring academic integrity through automated online exam proctoring a decade long systematic review (Discover Education, Springer) **(13 Feb, 2026)**
  - 45/80 papers from 2021-2023.
  - 80 papers studied from 2014-2024.
  - Only talks ML, DL, Privacy, Regulations (GDPR), False Positives, Dataset problems

    - Proves this topic is hot after COVID, only seen from AI perspective
    - Link: [Ensuring academic integrity through automated online exam proctoring a decade long systematic review | Discover Education | Springer Nature Link]

- Online Exam Proctoring: A Comprehensive Review and Critical Analysis (Asian Journal of Advanced Academic Research and Analysis) **(April 2025)**
    - Same focus, face recognition, gaze tracking, impersonation, ethical concerns.
    - Link: [(PDF) Online Exam Proctoring: A Comprehensive Review and Critical Analysis]

- Digital proctoring in higher education: a systematic literature review (Emerald) **(July 2023)**
    - Webcam, Microphone, Screen activity monitoring.
    - Concludes that digital proctoring is not perfect.
    - Link: [Digital proctoring in higher education: a systematic literature review - ScienceDirect]

- A Systematic-Narrative Review of Online Proctoring Systems and a Case for Open Standards (Open Praxis) **(August 2025)**
    - Same thing, no mention of VMs as such.
    - Link: [A Systematic-Narrative Review of Online Proctoring Systems and a Case for Open Standards | Open Praxis]

These are general peer-reviewed review papers, not some research papers targeting one specific angle in a research space.

My POV: VM detection is just a feature and its usuall a checklist by enterprise companies making such softwares. Also, it's quite a new thing and maybe this area/domain is yet not that important to cybersecurity professionals that they dig it up.

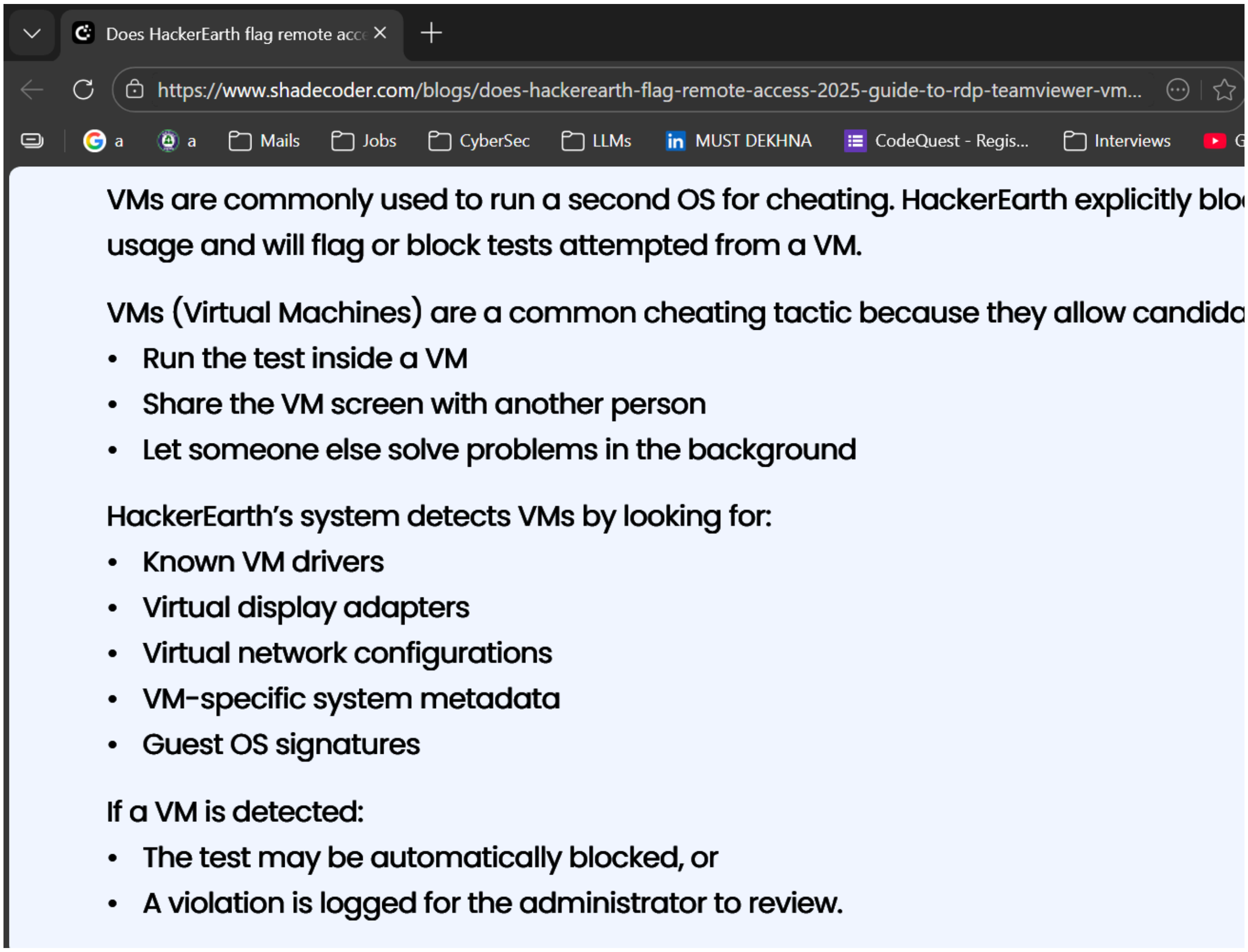
Simple VM Detection Techniques:

1) mac address
2) registry editor (LOCALMACHINE/SYSTEM/VBOX OR VMWARE)
3) vm helper processes names(e.g., sharing mouse, clipboard, or display)
4) vm specific drivers names, VM name
5) bios identifiers
6) disk/hard drives model names
7) CPU Flag Check

They can all be spoofed. How..? See below

Take this article of HackerRank cheating detection tactics [Does HackerEarth flag remote access? 2025 Guide to RDP, TeamViewer, VM rules and more | Shadecoder]

See this snapshot from this article:



VMs are commonly used to run a second OS for cheating. HackerEarth explicitly blo usage and will flag or block tests attempted from a VM.

VMs (Virtual Machines) are a common cheating tactic because they allow candida

- Run the test inside a VM
- Share the VM screen with another person
- Let someone else solve problems in the background

HackerEarth's system detects VMs by looking for:

- Known VM drivers
- Virtual display adapters
- Virtual network configurations
- VM-specific system metadata
- Guest OS signatures

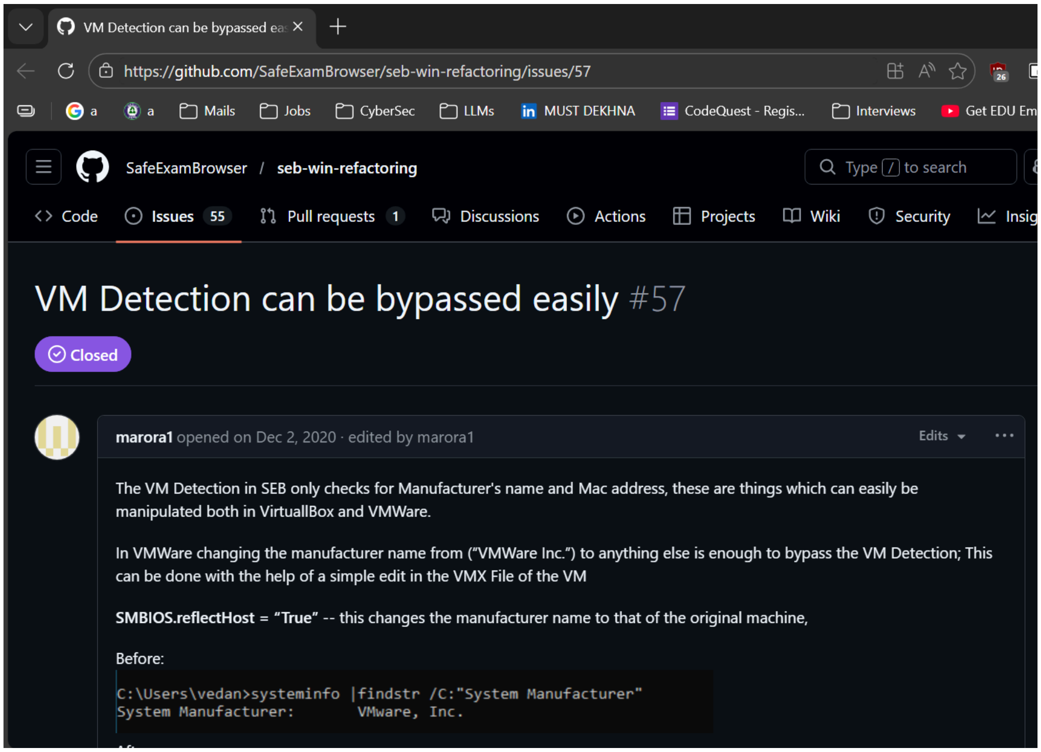
If a VM is detected:

- The test may be automatically blocked, or
- A violation is logged for the administrator to review.

These all can be spoofed. My research goes a step further and focuses on things that are almost impossible to spoof.

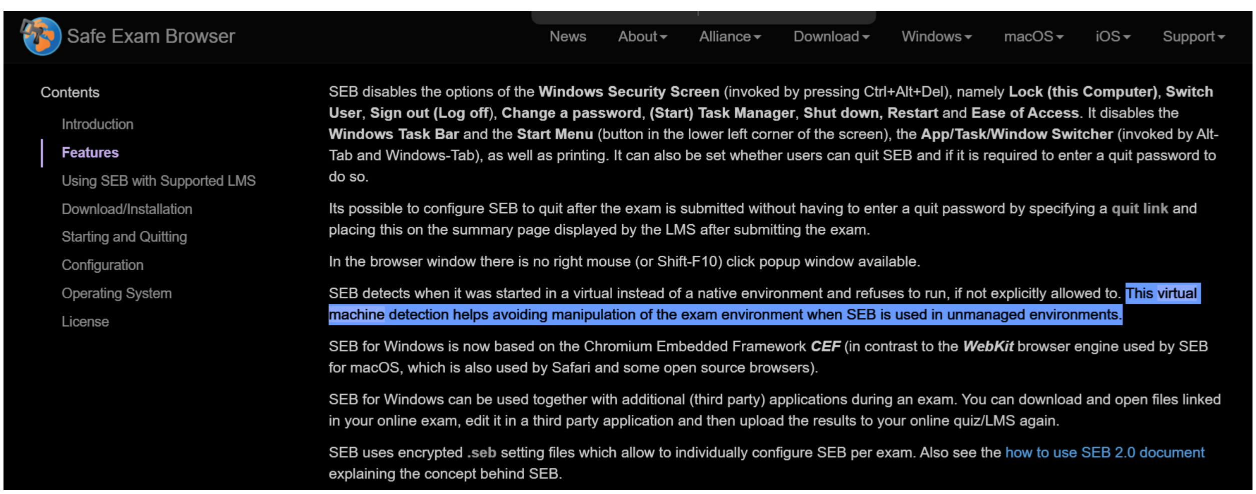
I am not saying by myself that normal VMs detection can be bypassed. See this link: [VM Detection can be bypassed easily · Issue #57 · SafeExamBrowser/seb-win-refactoring]

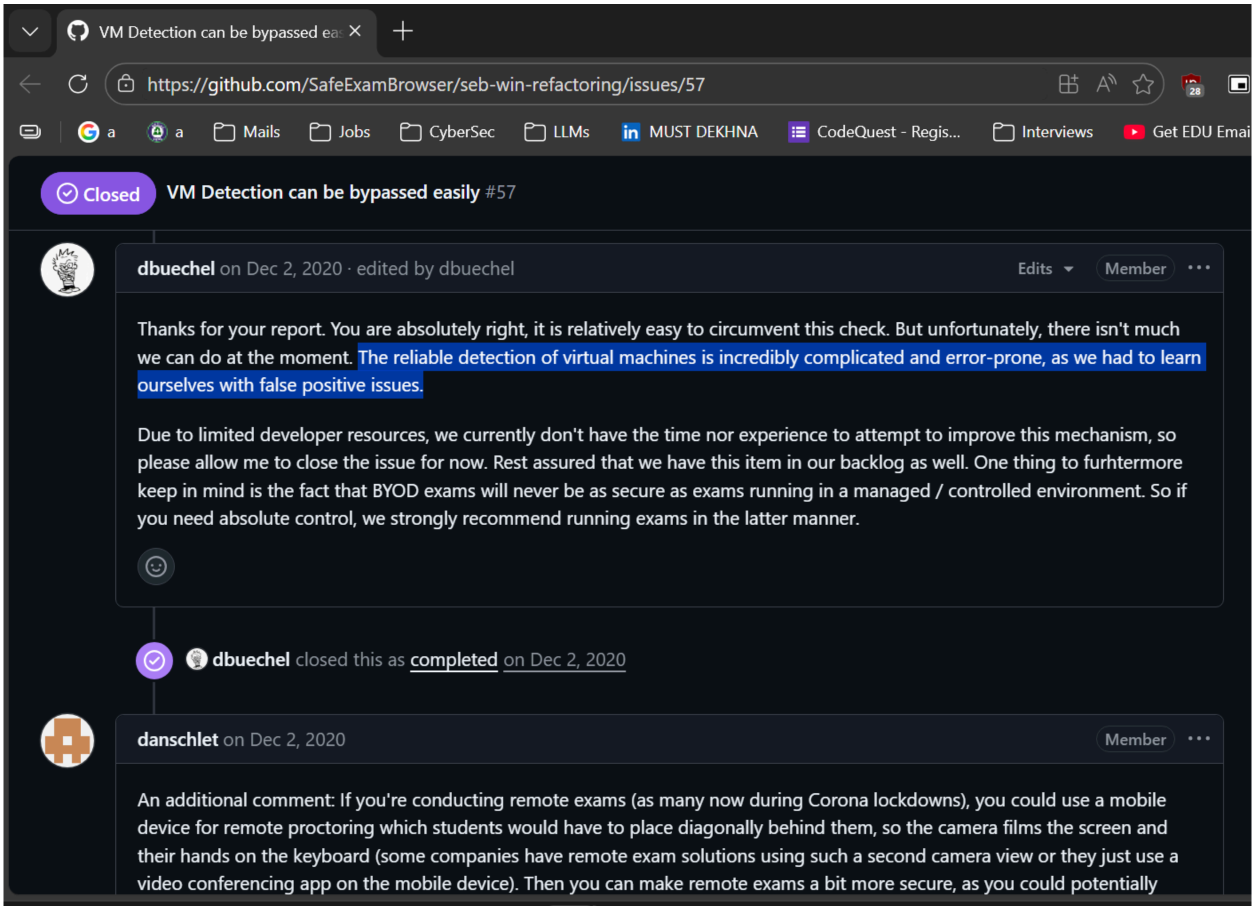
A snapshot of this link is below:



This is a well-known open-source exam-taking software. In it, an issue has opened (now closed) stating that normal VM detection can easily be bypassed.

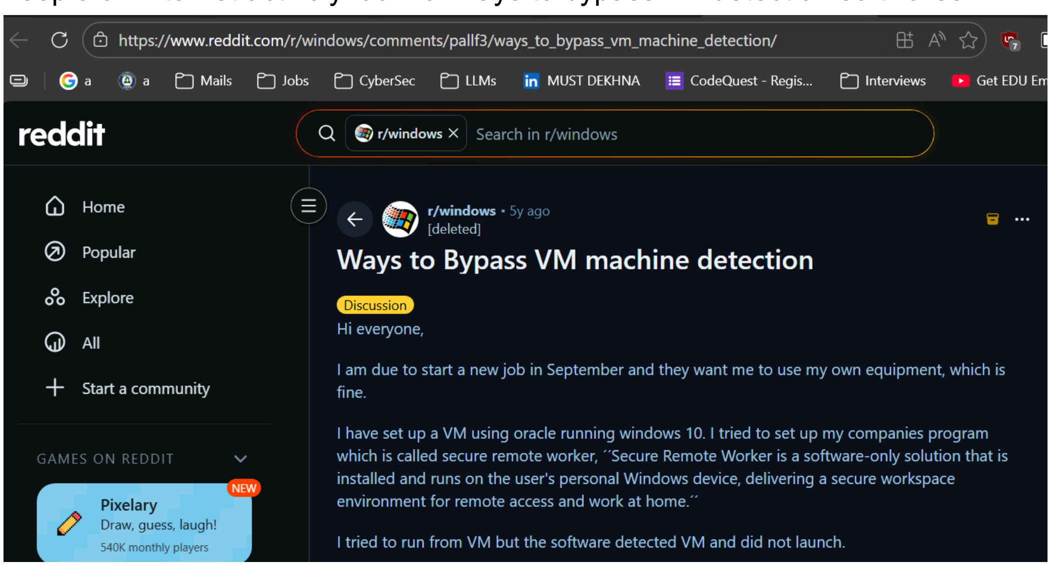
VM Detection is incredibly complex and has lots of false positives. [Safe Exam Browser - exam-taking application]

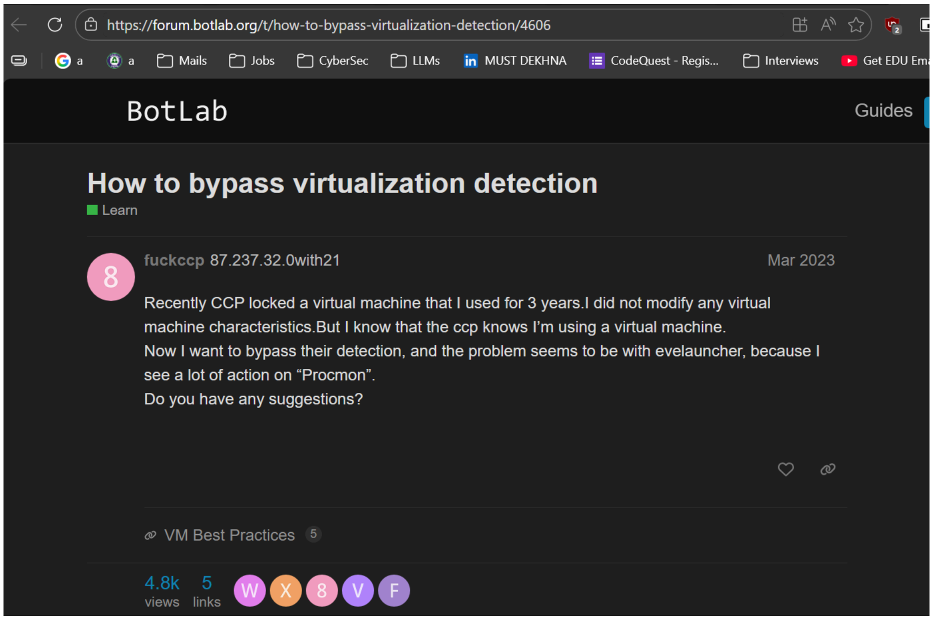




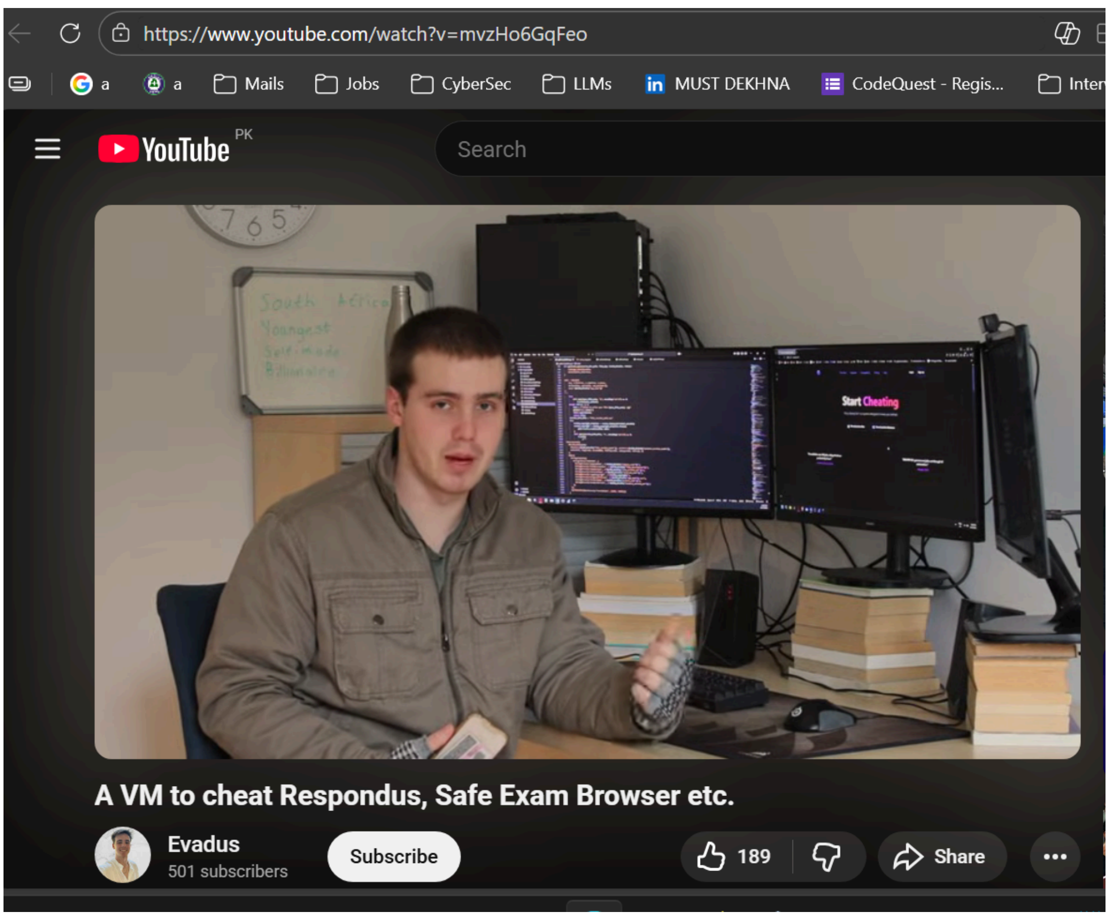
Virtual Machine Detection is already a major issue.

People on internet actively look for ways to bypass vm detection softwares.

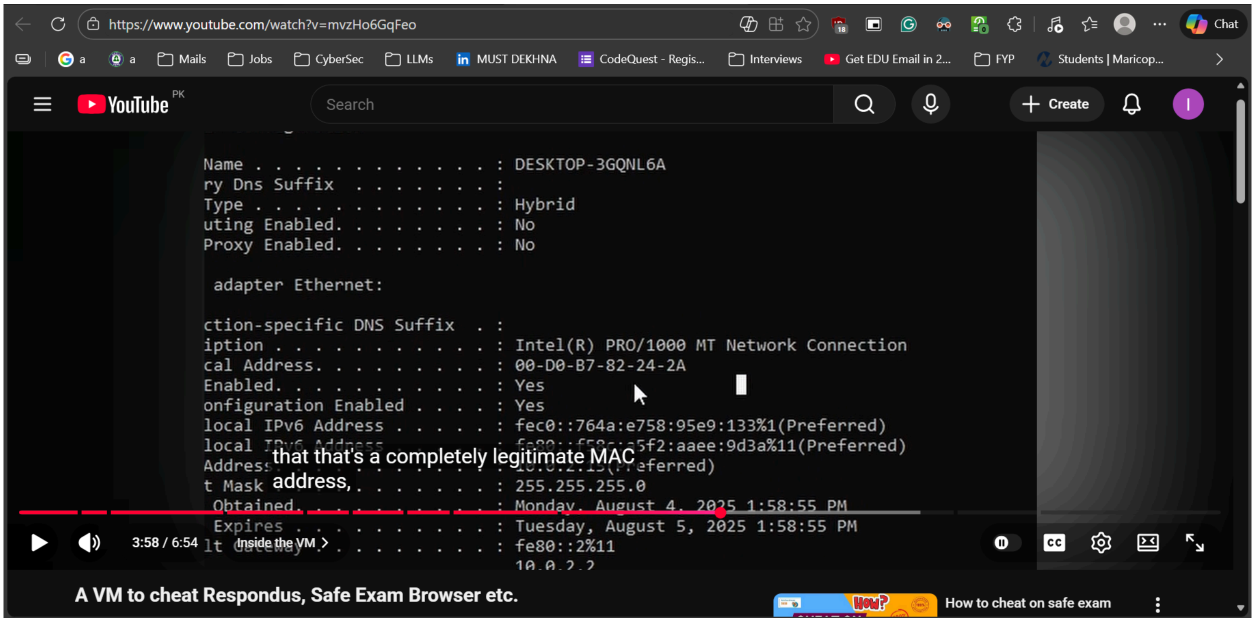




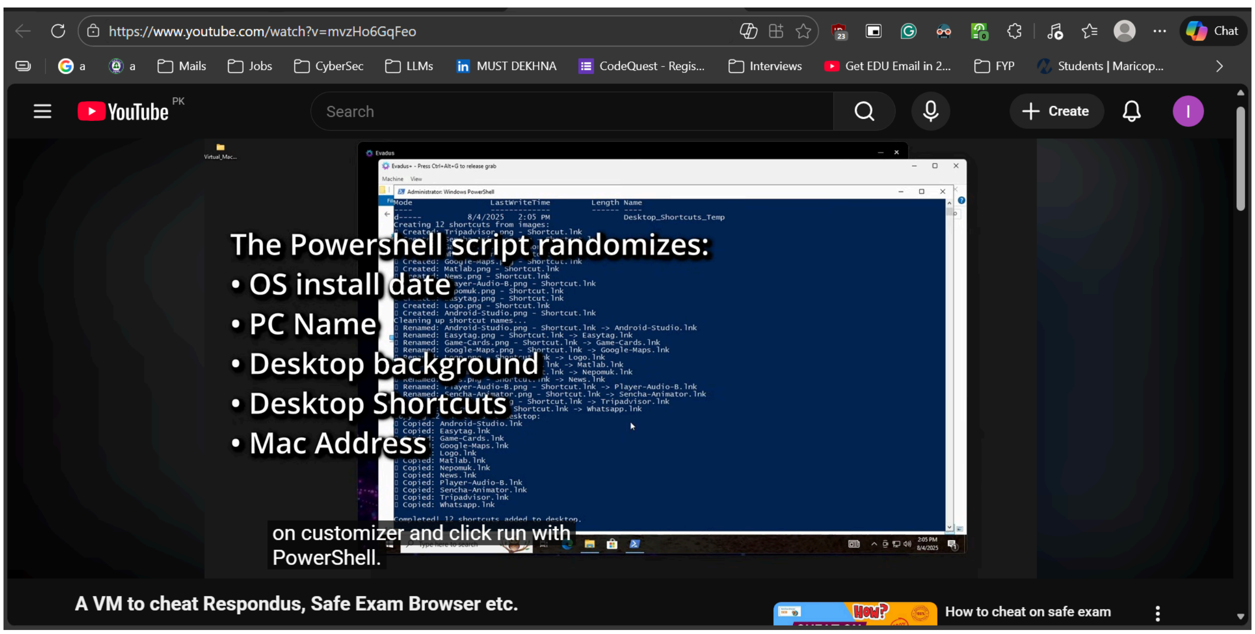
See this YouTube video: [A VM to cheat Respondus, Safe Exam Browser etc.]



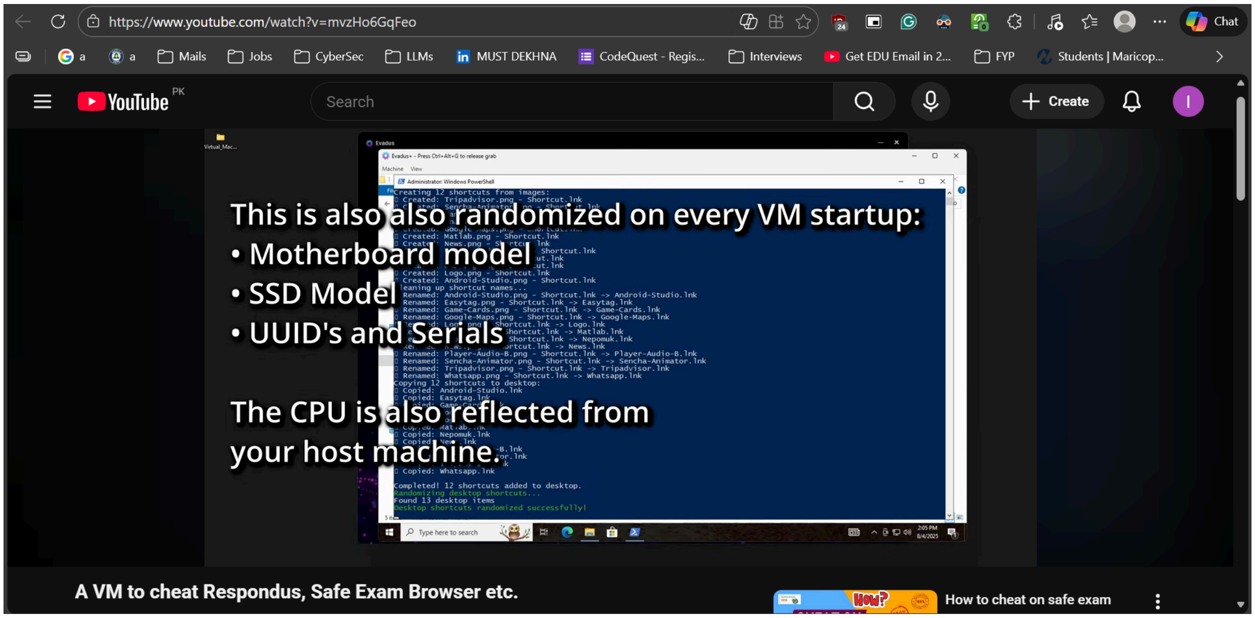
He shows how you can evade a exam-taking software VM Detection check. Below screenshot are from his video.



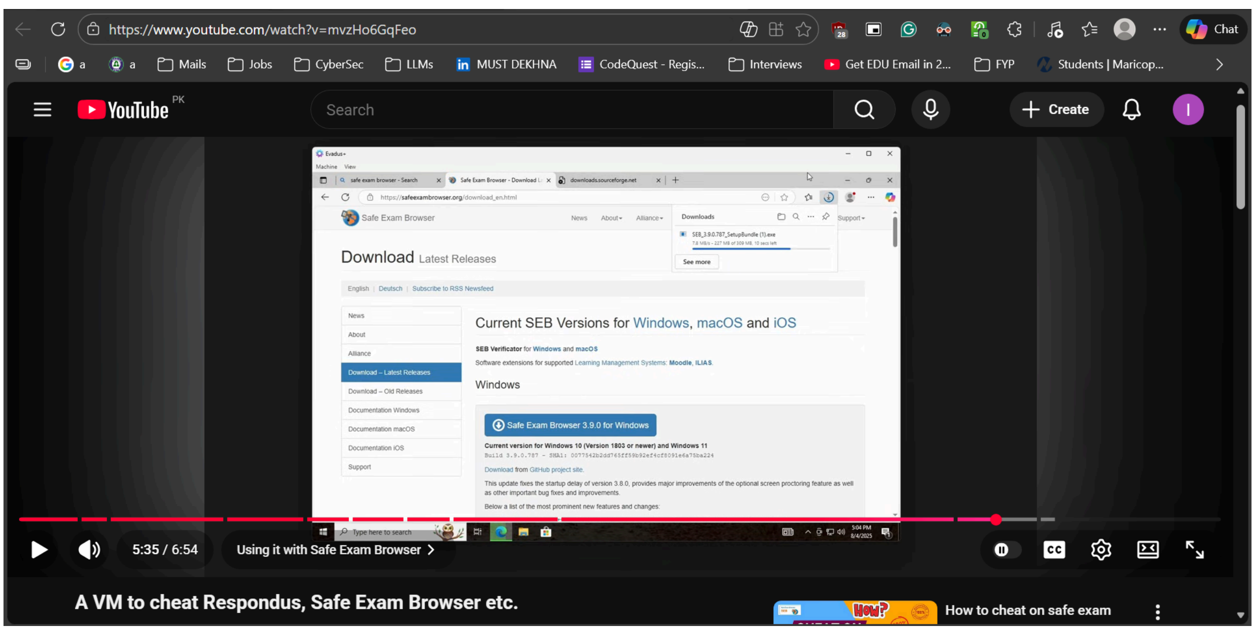
He’s showing his VM has a legitimate MAC address. Cant be detected now.



A script spoofing all VM things as I said above.

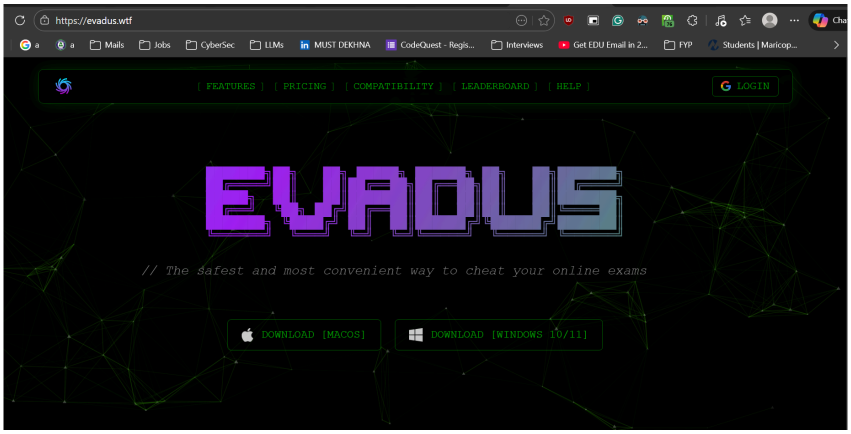


So, now its a normal VM that looks like a physical machine and a normal exam-taking application can never detect it.



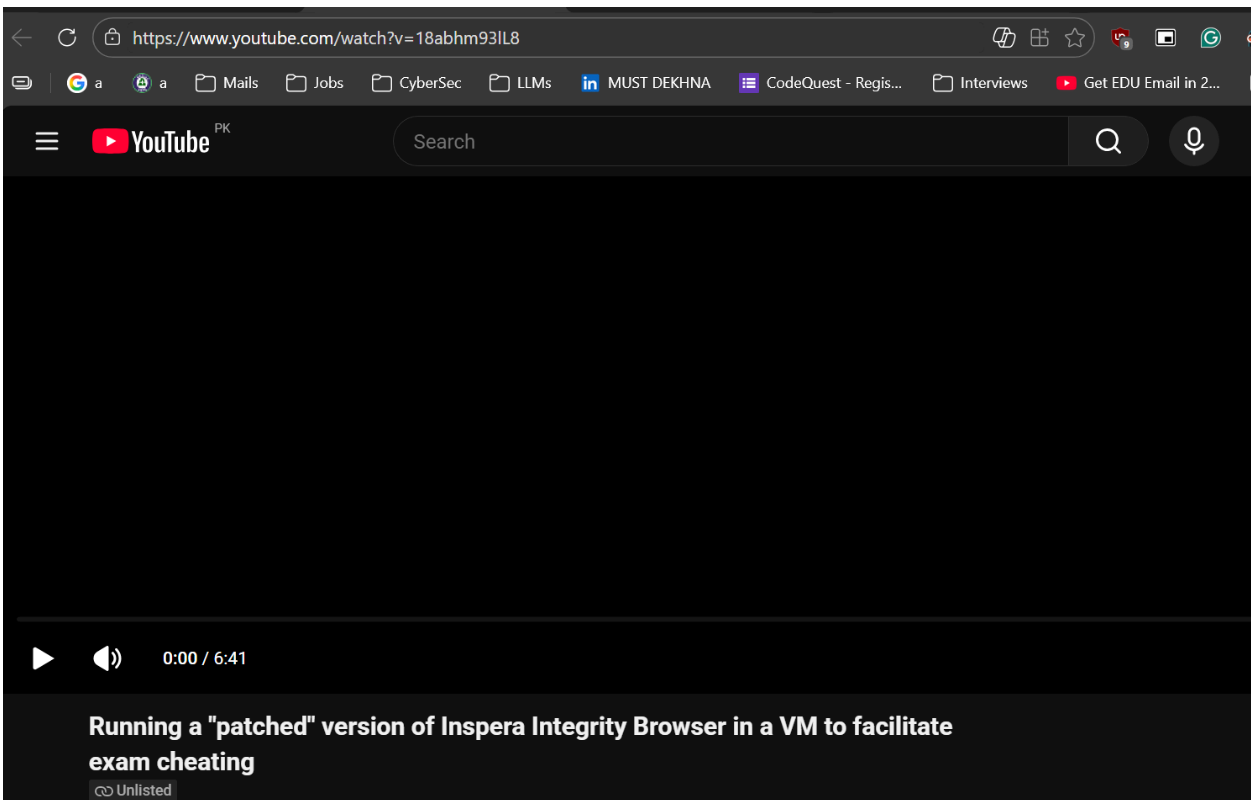
He’s testing Safe Exam browser and proving that spoofing a normal VM works.

He has just spoofed the PC Name, Driver names, some processes to match them to normal windows-based names and HOORAH!



These anti-cheating VMs rely on spoofing VMs characteristics.

Cheating Facilitation videos on YouTube: [Running a "patched" version of Inspera Integrity Browser in a VM to facilitate exam cheating]



Read this, how easy is it to cheat using virtual machines. [How networking technology enables cheating in online exams | APNIC Blog]
This blog is written by a professor from University of Auckland.

Only found one research paper of VM malpractices in exam-taking softwares.

([16-10604.pdf]) [A Virtual Machine-based e-Malpractice Mitigation Strategy in e-Assessment using System Resources and Machine Learning Techniques] (July 2024)

This paper combines AI alongside the visible normal virtual machine detection checks.
Calls for this future work:

examination systems that ensure fairness and uphold academic integrity.

It is, however, suggested that future research should:

- Focus on refining the model, expanding the dataset, and conducting real-world experiments to further validate and enhance the proposed approach.
- Additionally, it is believed and suggested that future research studies should show the impact of the research in the practical field.

Author contributions

THE END!

My 3 proposed techniques:

1. CPU Timing Attack (RDTSC):

You force the CPU to run a million tiny math problems.
You measure the time using a high-precision clock (RDTSC instruction).
On a real machine, the execution time is fast and consistent. On a VM, the time fluctuates (high "jitter") because the hypervisor has to intercept and translate the instructions, creating a measurable delay.

2. Thermal Zone Analysis:

You record the temperature. You run a heavy loop. You check the temperature again.
A real computer follows the laws of physics—it gets hotter when it works hard. A VM usually reports a "flat" or static temperature because it cannot access the physical thermal sensors of the host machine.

3. Hardware Artefact Scanning:

You check deep details like "SMBIOS Serial Number" or "Cache Line Size."

Real PC manufacturers (like Dell or HP) always populate these fields. Virtual Machines often leave them blank, generic, or set to default values like "0" or "To Be Filled By O.E.M.", which is a clear sign of virtualization.

DONE!